

# Environmental decision support: rare events, monitoring and inversion, and uncertainty quantification

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Responsible research, Birmingham, 2026  
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# Acknowledgement and overview

## Acknowledgement

- Cambridge, Durham, Lancaster, Melbourne, Shell

## Overview

- This is a talk about responsible research in environmental decision support
- My take on “responsible research”
- Remote sensing (airborne, line-of-sight, satellite)
- Extremes (non-stationary marginal, multivariate)

# Responsible research

## What is it?

- Research which genuinely tries to leave to world and everyone and everything it touches in a better state
- Is scientifically rigorous
- Avoids harm to people, animals and environment
- Serves the public good

# Responsible research

## In academia?

- Addresses “real-world societal needs”
  - Not “chasing citations” or “top journals” or “oneupmanship”
  - Not transient “flavour of the month” topics
  - (But what about “research for research’s sake” and unanticipated future impact?)
- Tries to manage “dual-use dilemmas”
  - e.g. Understanding of atomic physics from early 1900s led to Hiroshima and Chernobyl; but it also led to unprecedented widespread and lasting public good
  - Where could AI lead?
- Provides “open-source” science
  - Transparent, accessible, and shared freely

# Responsible research

## In heavy industry and fossil fuel sectors?

- Mitigates immediate harm
  - Methane detection and abatement
  - Safe decommissioning of large-scale facilities
- Mitigates medium-term harm: transforms industrial processes
  - Electricity in place of hydrocarbons as the energy source for heat-intensive reactions (e.g. needed to create plastics, chemicals, fertilizers, extract lithium from rock)
  - Depleted oil and gas reservoirs for carbon capture and storage
  - Existing pipeline infrastructure to carry green hydrogen in place of methane
  - Alternative sources of energy: renewables, small modular nuclear reactors?
- Transforms society
  - "Addressing the economic reality of energy-dependent communities and wider society"
- Green-washing
  - Who's the judge?

# Responsible research

## Reality check!

- Is responsible research compatible with a free-market economy?
  - Driven by “profit motive” and competitive “advantage”
    - At a corporate and individual level
  - Fundamentally misaligned with RR (driven by “public good”)
  - From a free-market perspective, RR is a cost, it reduces profit and competitive advantage  $\Rightarrow$  it's suboptimal (to say the least) to do it
- Public pressure is ultimately the handbrake
  - Public demand for regulation and other “guardrails”
  - Public demand (and resulting sources of investment) for ethical processes and products
  - Education, transparency, open science
- But does the handbrake really work?
  - Think about climate change
    - Societal inequality is increasing (e.g. in UK since late 1970s) <https://wir2026.wid.world/>
    - The wealthiest in society behave most irresponsibly regarding climate change  
[https://wid.world/www-site/uploads/2025/10/Climate\\_Inequality\\_Report\\_2025\\_Final.pdf](https://wid.world/www-site/uploads/2025/10/Climate_Inequality_Report_2025_Final.pdf)
  - Think about ill-informed or even malicious public pressure

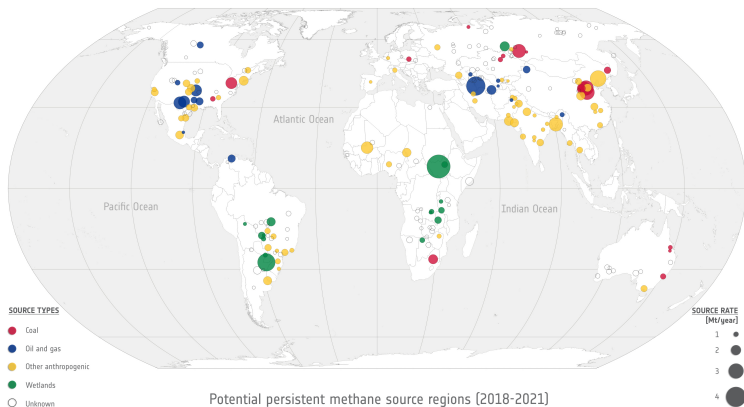
# Remote sensing of gaseous emissions

# Remote sensing: why is this important?

- Greenhouse gases cause global warming with potentially catastrophic effects;  $\text{CO}_2$  and  $\text{CH}_4$  are the two most important greenhouse gases
- “Natural”  $\text{CO}_2$  sources (aerobic respiration, organic decomposition, ocean-atmosphere exchange, volcanoes, wildfires), as well as “anthropogenic” sources (e.g. combustion, which we can try to reduce)
- “Natural”  $\text{CH}_4$  sources (anaerobes in wetlands, ruminants, permafrost; ocean hydrates; wildfires), as well as “anthropogenic” sources
- $\text{CO}_2$  stays in the atmosphere for centuries;  $\text{CH}_4$  for  $\approx 12$  years only
- $\text{CO}_2$  has a global warming potential of unity (by definition);  $\text{GWP}(\text{CH}_4 \text{ over } 20 \text{ years}) \approx 80$  (much worse!)
- $\text{CH}_4$  emissions responsible for **half** the extreme heat absorbed by the atmosphere since the industrial revolution
- If we can stop  $\text{CH}_4$  getting into the atmosphere, we can provide “immediate mitigation”
- We can look for  $\text{CH}_4$  sources locally (using point sensors, line-of-sight sensors or drones) or on a global scale using satellites



# Remote sensing: super-emitters



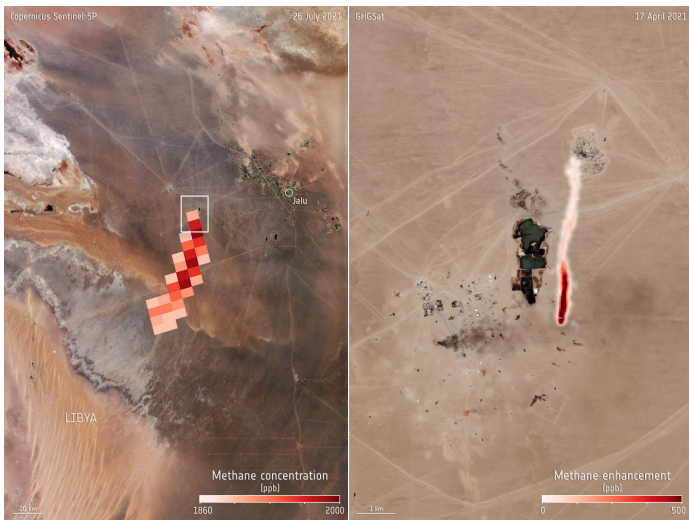
[https://www.esa.int/Applications/Observing\\_the\\_Earth/Copernicus/Sentinel-5P/Top\\_10\\_persistent\\_methane\\_sources](https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-5P/Top_10_persistent_methane_sources) using data from the Copernicus Sentinel-5P TROPOMI satellite

# Remote sensing: fracking in the Permian basin (Texas)



<https://fossilfuel.com/the-west-texas-oil-boom-that-changed-the-world/>

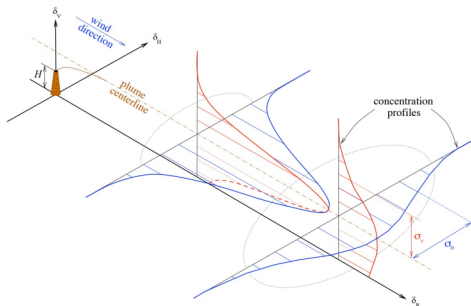
# Remote sensing: super-emitter plumes (Lybia)



<https://phys.org/news/2023-09-trio-sentinel-satellites-methane-super-emitters.html>

# Remote sensing: key ideas

- An unknown source of gas
- Gas transported on the wind
- We observed the gas remotely using a sensor
- Given the remote measurements and some knowledge of the wind field, can we characterise the source (size and location)?



Stockie [2011]

# Remote sensing: formulation

- Statistically, we have a problem of probabilistic inversion

$$y = As + b$$

- Source strength  $s(q)$  at fixed location  $q$
- Remote measurements  $y(\delta_t)$  at locations  $\delta_t$  (relative to source location) in time
- Background  $b(\delta_t)$  at the measurement location in time
- Coupling matrix  $A(\delta_t, \sigma)$  in time (e.g. Gaussian plume) between source the measurement locations, with (potentially unknown) coupling parameters  $\sigma$
- Spatio-temporal wind field  $w(x, t)$  in space and time

# Remote sensing: methodology and contributions

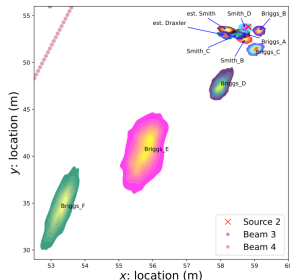
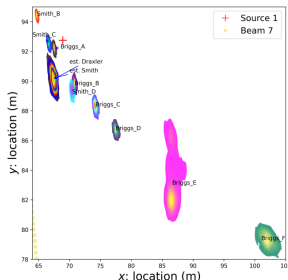
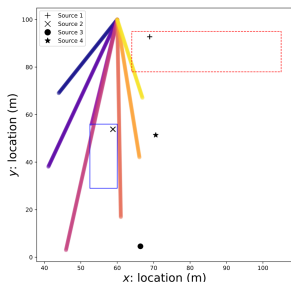
## Methodology

- MCMC, full conditionals where possible
- MH with gradient-based (mMALA) proposals otherwise
- More recently, MLP

## Contributions

- MCMC quantification of land-fill emissions: Hirst et al. [2013]
- Line-of-sight inversion: Hirst et al. [2017]
- Permian basin (satellite): Roberts et al. [2022]
- MCMC for estimation of coupling parameters: Newman et al. [2025a]
- Deep learning for inversion: Newman et al. [2025b]

# Remote sensing: illustrative results



Newman et al. [2025a]. “est” are source estimates with opening angles estimated as part of the MCMC. Other estimates make explicit assumptions (often guessed) about opening angles.

# Remote sensing: what next?

- Remote sensing (and associated inference) rapidly growing field
- SRON: <https://www.sron.nl/en/pillars/science/earth/methane/>
- GHGSat: <https://www.ghgsat.com/>
  
- Evolving regulatory frameworks
- Satellites, line of sight and point sensors
- Independent verification of industrial reporting of greenhouse gas emissions
  
- “Earth observation”
- <https://www.gov.uk/government/collections/earth-observation-eo>
- <https://www.earthdata.nasa.gov/learn/earth-observation-data-basics>
- <https://earth.esa.int/eogateway>



# Characterising extreme ocean environments

# Extreme environments: why is this important?

- Extreme environments are **dangerous to life** and human infrastructure
  - Extreme ocean storms, earthquakes, floods, wildfires, volcanos, ...
  - Want to build processes and structures that are “safe” given an inherently “risky” environment
  - e.g. “This structure must withstand the 1-in-1000-year event”
- Influence and improve **existing regulatory environment**
  - Historically, lack of data and methodology for good risk analysis, so existing industry standards use “sub-optimal” approaches
  - Modern methods are **more** complete and coherent (in terms of physical and statistical understanding)
  - But people must be **convinced** to switch to better approaches
- Combine best new ideas from academia with best practice from practical experience

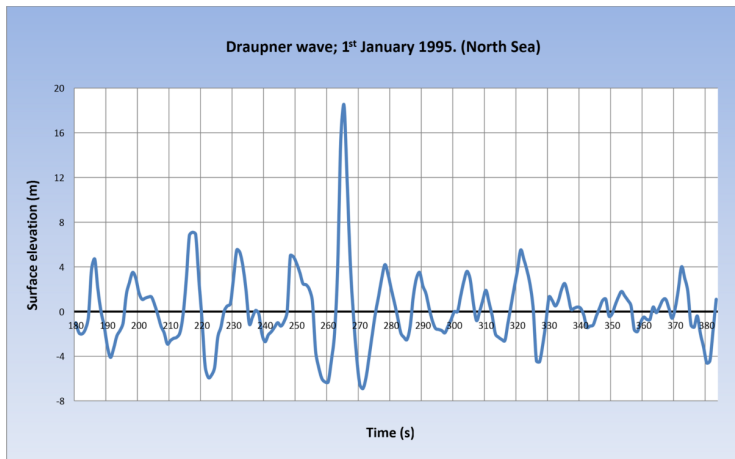
# Extreme waves: spectacular scale



Offshore Portugal, 24m wave height, November 2017 (The Guardian)

- Nazaré is a great source of huge coastal waves

# Extreme waves: spectacular scale



Laser readings, 1 January 1995. Wave 25.6m, crest 18.5m (Statoil / Equinor)

- Maximum recorded wave height > 30m (multiple events, various sources)
- Maximum recorded significant wave height : 19.0m (buoy, North Atlantic, 4 Feb 2013, WMO)

# Extreme waves: impact damage



Norwegian Dream, Atlantic, 2007  
(gcaptain.com)



Ike, Gulf of Mexico, 2008  
(Joe Richard)

# Extreme environments: non-stationary marginal

- Environmental extremes of (univariate)  $Y$  vary continuously with multidimensional covariates  $\Omega$
- Asymptotic theory gives form of distribution of exceedances of high threshold  $\psi$

$Y|(\Omega, Y > \psi) \sim GP(\xi, \sigma, \psi)$ , generalised Pareto with parameters  $\xi, \sigma, \psi$

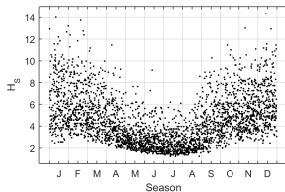
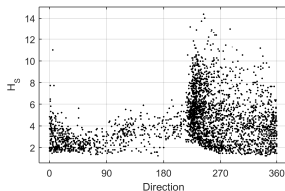
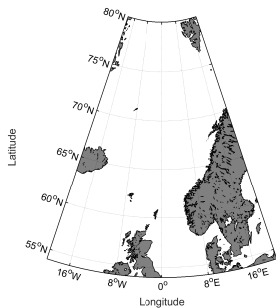
- Inferences should reflect sources of uncertainty fairly
- Need statistical and computational efficiency
- **Predict extreme quantiles** of  $Y$
- **Assess risk** (or expected loss  $\mathbb{E}(L)$ ) for system  $S = s_0$  due to  $Y$  and structural response  $R$

$$\mathbb{E}(L|S = s_0) = \int_r \int_y \int_\omega L(r|S = s_0) f_{R|Y}(r|y) f_{Y|\Omega}(y|\omega) f_\Omega(\omega) d\omega dy dr$$

- Use cases: Offshore and coastal design, weather windows and alerts
- Jones et al. [2018], Hansen et al. [2020], Towe et al. [2021]

# Directional-seasonal: application

Storm peak significant wave height at northern North Sea location; clear directional and seasonal variability in storm severity; directional variability more dramatic at around 225°; seasonal variability more gradual.



# Directional-seasonal: the model

## Density

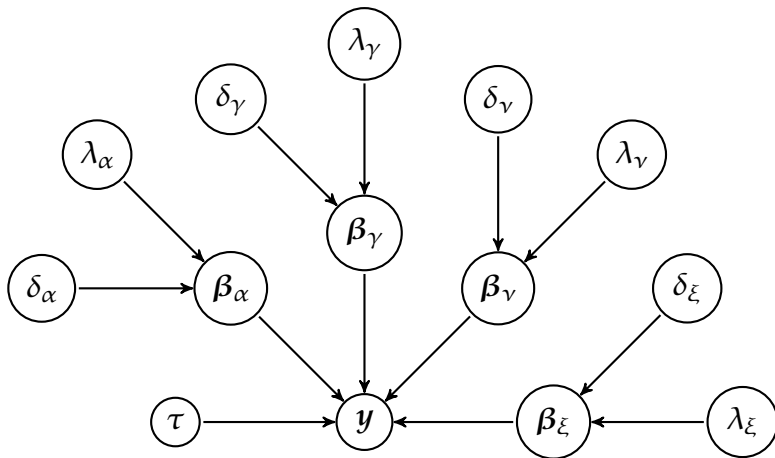
- $f(y|\xi, \sigma, \alpha, \gamma, \psi, \tau) = \begin{cases} \tau \times f_{TW}(y|\alpha, \gamma) & \text{for } y \leq \psi, \text{ a truncated Weibull (or similar)} \\ (1 - \tau) \times f_{GP}(y|\xi, \sigma) & \text{for } y > \psi \end{cases}$
- Threshold non-exceedance probability  $\tau$  to be inferred
- Physics suggests parameters  $\alpha, \beta, \rho, \xi, \sigma, \psi$  and  $\tau$  vary smoothly with covariates  $\theta, \phi$
- Randell et al. [2016]

## Covariate representation: B-splines

- Values of  $\eta \in \{\alpha, \beta, \rho, \xi, \sigma, \psi, \tau\}$  on some index set of covariates take the form  $\eta = B\beta_\eta$
- $B$  takes the form  $B_\phi \otimes B_\theta$ , GLAMs provide efficient manipulation
- Spline roughness penalty is **quadratic form**  $\beta_\eta' P_\eta \beta_\eta \Rightarrow$  motivates prior for  $\beta_\eta$
- $P_\eta = \lambda_{\eta\theta} P_{\eta\theta} + \lambda_{\eta\phi} P_{\eta\phi}$ , includes **stochastic roughness penalties**  $\{\delta_{\eta\theta}, \delta_{\eta\phi}\}$
- Brezger and Lang [2006], Currie et al. [2006], Eilers and Marx [2010], Zanini et al. [2020]



## Directional-seasonal: DAG for size of threshold exceedances

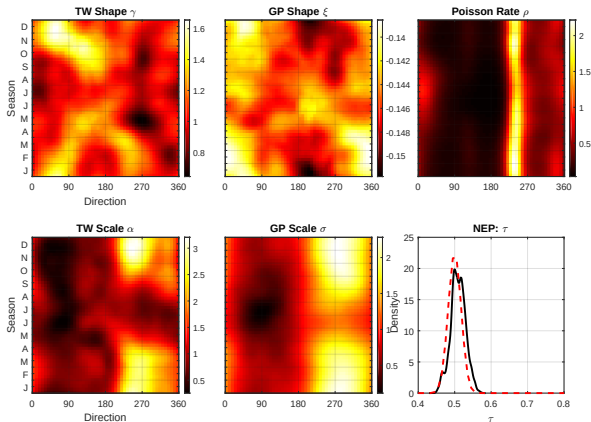


# Directional-seasonal: inference

- Sampling from full conditionals
- Gibbs sampling when full conditionals available in closed form
- Metropolis-Hastings (MH) within Gibbs otherwise, using suitable proposal mechanisms (mMALA)
- Roberts and Stramer [2002], Girolami and Calderhead [2011], Xifara et al. [2014]

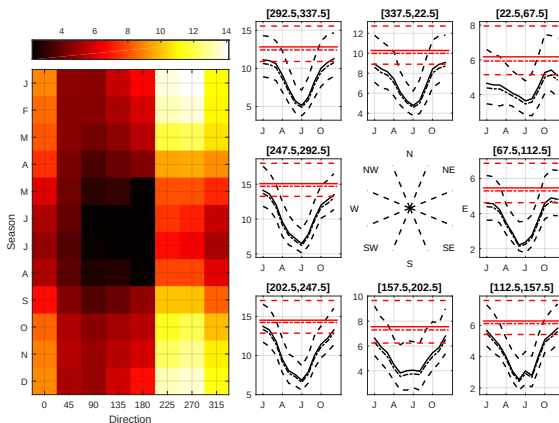
# Directional-seasonal: parameter estimates

Prior for  $\tau$  in red, posterior in black; all parameters except  $\xi$  and  $\tau$  suggest strong directional variation; seasonal variation less pronounced but clear for  $\alpha$ ,  $\sigma$  (and  $\rho$ );  $\xi$  effectively constant; sample not informative about  $\tau$ .



# Directional-seasonal: return values

Predictive distribution of the 100-year maximum (in metres); directional and seasonal variability of the median estimate on lhs; seasonal variation of predictive distribution for directional octants (2.5%, 37%, median and 97.5% values) in black; corresponding omni-seasonal estimates in red; large difference between S and SW; smooth seasonal variation.



# Extreme environments: the future

## Use case

- Risk analysis for **multivariate spatio-temporal** processes

## Models

- Max-stable processes, copulas, conditional extremes, SPAR, engineering models
  - Ross et al. [2017], Tendijsck et al. [2019], Tendijsck et al. [2023], Murphy-Barltrop et al. [2024], Sando et al. [2024]
- **Key idea:** dependence **in body** and dependence **in tail** are **different**  
 $[X, Y] \sim N(0, [1 \ \rho; \rho \ 1]), \rho < 1, \lim_{x \rightarrow \infty} \Pr(Y > x | X > x) = 0$

## Risk assessment

- Expected loss  $\mathbb{E}(L)$  for system  $S = s_0$  due to **multivariate**  $Y$  and **multivariate** structural response  $R$

$$\mathbb{E}(L|S = s_0) = \int_r \int_y \int_\omega L(r|S = s_0) f_{R|Y}(r|y) f_{Y|\Omega}(y|\omega) f_\Omega(\omega) d\omega dy dr$$

# Summary

# Responsible research in environmental decision support: summary

## Scientific rigour

- In my experience of “conventional applied mathematical sciences”, it’s genuine unintended lack of insight that leads to lack of scientific rigour: “cock-up” rather than “conspiracy”
- But wilful ignorance is no defence (e.g. with respect to use of LLMs)

## Avoid harm

- Usually, in “conventional applied science”, it’s genuine unintended lack of insight that leads to (potential) harm
- At the industrial “coal face”, my impression is that the profit motive and competitive advantage can sometimes be overwhelming (or “blinding”)

## Public good

- Ultimately subjective
- The future is unknown
- But these are no reason to stop us **trying** to behave rationally and with a clear conscience

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